Warsaw University of Technology
Faculty of Mathematics
and Information Science

Master's diploma thesis

in the field of study Computer Science and Information Systems
and specialisation Artificial Intelligence

Multi-model LLM system with dynamic planning and model selection optimization

Jakub Knyspel

student record book number 313282

thesis supervisor
dr hab. Maria Ganzha, prof. nzw.

WARSAW 2026

Abstract

Multi-model LLM system with dynamic planning and model selection optimization

TODO

Keywords: TODO, ...

Streszczenie

Implementacja wielomodelowego systemu LLM z dynamicznym planowaniem i optymalizacją doboru modeli

TODO

Słowa kluczowe: TODO, ...

Contents

1. Introduction **11**

2. Related Work **12**

2.1. RouteLLM 12

2.2. RoRF - Routing over Random Forests 12

3. System Requirements & Constraints **13**

4. Description of the System **14**

5. Scoring **15**

5.1. Dataset Selection 15

5.2. Features 16

5.3. Data Analysis 16

5.4. Prompt-invariant Scoring 16

5.5. LLM Embedding 16

5.6. Features 16

5.7. Metrics 17

5.7.1. DNN-based Embedding 17

5.7.2. Tuning-based Embedding 17

5.7.3. Attention Embedding 17

5.7.4. Prompt-invariant Embedding-based Models 17

5.8. Simple Models 17

5.8.1. DNN Model 17

5.8.2. Gradient Boosting Model 17

5.9. Complex Models 17

5.9.1. Transformer Model 17

5.9.2. Response Predictive Model 17

5.9.3. Multistage Models 18

6. Response Length Prediction **19**

6.1. Input and Output Data 19

6.2. Models 19
6.2.1. Gradient Boosting Model 19
6.2.2. DNN Model 19
7. Analysis of the Solution 20
8. Conclusion 21

1. Introduction

TODO

2. Related Work

2.1. RouteLLM

TODO

2.2. RoRF - Routing over Random Forests

TODO

TODO: More references

3. System Requirements & Constraints

TODO: What did I decide to do? What constraints did I settle on?

4. Description of the System

TODO: Description of system components, communication, etc. TODO: Also include model selection based on price and score.

5. Scoring

TODO

5.1. Dataset Selection

The title of the most popular human-evaluated LLM preference dataset without a doubt belongs to *Chatbot Arena Conversations* [TODO: ref]. It is used as primary human-evaluated dataset in referenced works [TODO: verify and ref].

Chatbot Arena Conversations data consists of a pair of responses from different LLM models to a single user prompt, along with a comparison label, determining which model's answer is considered better by the user. When comparison is unclear, the label can also indicate that both responses were sufficient for a given query, or that neither model managed to produce a good answer. Selection of LLMs is sufficiently wide for a model to learn a broad spectrum of differences between different LLM models, while prompts span a variety of topics.

However, there is a crucial issue: published in 2022 [TODO: verify], *Chatbot Arena Conversations* dataset only includes comparisons between rather old LLM models. This isn't a disqualifying problem, as my work doesn't focus on particular LLM models. Though, A decision to use this dataset would require two important implications: that human preference during evaluation of a pair of responses remains unchanged, and that the general distribution of responses produced by LLMs is similar.

It's hard to evaluate how close to the truth the first assumption is. While human preferences likely did not change drastically over the last four years, expectations of what LLMs should be able to do did undergo an undeniable change, simply due to their capabilities increasing. Another significant difference may be caused by a wider adoption - people whose responses were used in *Chatbot Arena Conversations* dataset belonged to a significantly narrower group of "early adopters" of LLM technology, compared to a group of LLM users today.

These reasons prompted me to decide to rely on a newer alternative to *Chatbot Arena Conversations* dataset. Collected and released by the same organization (*LMSYS* [TODO: ref]),

LMArena Human Preference is a successor of aforementioned dataset. It's structure closely resembled *Chatbot Arena Conversations*, also including human comparison results over a pair of LLM responses to the same prompt. Additionally, the dataset includes a range of common prompt labels [TODO: describe], which can be used as additional features in any prediction task. *LMArena Human Preference* is also an improvement in terms of size. It contains around 140 thousand comparison pairs (compared to *Chatbot Arena Conversations*'s 33 thousand), and includes responses from 53 different models (compared to *Chatbot Arena Conversations*'s 20).

It's important to note that due to a very similar structure, these datasets can be used interchangeably. Due to previously mentioned reasons, mainly *LMArena Human Preference* was used for model training and evaluation throughout this work. However, comparisons between datasets are made where relevant, highlighting important differences between their data.

5.2. Features

TODO

5.3. Data Analysis

TODO

5.4. Prompt-invariant Scoring

TODO

5.5. LLM Embedding

TODO

5.6. Features

TODO

5.7. Metrics

TODO

5.7.1. DNN-based Embedding

TODO

5.7.2. Tuning-based Embedding

TODO

5.7.3. Attention Embedding

TODO

5.7.4. Prompt-invariant Embedding-based Models

TODO

5.8. Simple Models

5.8.1. DNN Model

TODO

5.8.2. Gradient Boosting Model

TODO

5.9. Complex Models

5.9.1. Transformer Model

TODO

5.9.2. Response Predictive Model

TODO

5.9.3. Multistage Models

TODO

6. Response Length Prediction

TODO: Should it be a separate chapter?

6.1. Input and Output Data

TODO

6.2. Models

6.2.1. Gradient Boosting Model

TODO

6.2.2. DNN Model

TODO

7. Analysis of the Solution

TODO: Tests of all parts of the system, etc.

8. Conclusion

TODO

List of symbols and abbreviations

nzw.	nadzwyczajny
*	star operator
~	tilde

If you don't need it, delete it.

List of Figures

If you don't need it, delete it.

Spis tabel

If you don't need it, delete it.

List of appendices

1. Appendix 1
2. Appendix 2
3. In case of no appendices, delete this part.